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Flood susceptibility mapping using multiple multi-criteria decision-making methods, Gaibandha district of northern Bangladesh

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Abstract

Flooding is a frequent and intense natural hazard in Bangladesh, particularly in river-dominated regions, exacerbated by the seasonal monsoon rains. Gaibandha, situated in the northern part of the country, is highly vulnerable to flooding, resulting in significant damage and loss of life and property every year. This study aims to identify flood-conditioning factors, local-level susceptibility, comparing multi-criteria decision-making (MCDM) models for determining the flood-prone areas and generating the flood susceptible map for Gaibandha district. This study investigates the key hydro-geomorphological factors that affect flood susceptibility in the data-scarce Gaibandha district, including elevation, slope, aspect, topographic wetness index, rainfall, curvature, drainage density, normalized difference vegetation index, distance from rivers, distance from road, land use/land cover, and topographic roughness index. All the factors are statistically significant, derived from multicollinearity and Pearson's correlation method. The study applied three MCDM techniques: analytical hierarchy process (AHP), analytic network process (ANP), and fuzzy AHP (FAHP), to map areas most susceptible to flooding. The findings revealed that 44–54% of the district is moderately susceptible to floods, 12–16% fell within the high-susceptibility zone, and only a small portion (0.7–1.1%) was classified as very highly susceptible. The predictive performance of the flood susceptibility maps produced by each method was assessed using the receiver operating characteristic curve. In the traditional approach, ANP achieved the highest accuracy, with an area under the curve of 0.885, marginally exceeding the results of FAHP (0.883) and AHP (0.881). In contrast, FAHP (0.739) performed slightly better than AHP (0.727) and ANP (0.727) in DeLong's test and bootstrapping, revealing the higher accuracy in both methods. Thus, ANP emerged as the most reliable approach for distinguishing flood-susceptible zones in the context of Gaibandha District and supported datasets, while FAHP depicts higher accuracy in DeLong's test and bootstrapping for the study area. This study provides an innovative and reliable approach for mapping flood susceptibility in Gaibandha and similar regions, offering valuable insights for flood susceptibility management. The results can guide policymakers in formulating effective mitigation strategies and enhancing community resilience to flooding.

Keywords Flood susceptibility, Multicollinearity, MCDM, AHP, ANP, FAHP, Gaibandha

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1 Introduction

Flooding has become one of the most frequent and devastating natural hazards in Bangladesh, particularly exacerbated by rapid climate change and global warming [1]. Over the last decade, hydro-meteorological extreme events have intensified, leading to catastrophic damage and frequent disasters [2–5]. As reported by the Emergency Events Database (EM-DAT), over 20 million people have been affected by such events, with the annual cost of damage surpassing US$350 billion globally [6–8]. Flooding, in particular, is a recurring disaster, especially in Asia, where tropical and subtropical regions face severe floods each year due to the seasonal monsoon rains, which account for approximately 80% of annual rainfall [9–11]. River bed sedimentation and river bank erosion due to water flow and channel migration decrease water bearing capacity of the river and exacerbate floods [12, 13]. Prolonged rainfall, combined with dynamic river discharge, leads to overflows and inundation in downstream regions like Bangladesh [14–16].

In Bangladesh, flooding is caused by a complex interaction of elements, including the country's geographical location at the outflow of the Ganges-Brahmaputra-Meghna (GBM) river system, its shallow riverbeds, inadequate drainage conditions, human activities, and the yearly summer monsoon [17–21]. These factors, along with its tropical climate, lead to annual inundation of about 20% of the country's total area [22]. Riverine, rainfed, and flash flooding occur frequently, leading to fatalities, infrastructure damage, livelihood disruption, and deterioration of public health, sanitation, and agricultural systems [23–26].

Historical flood events in Bangladesh, including those in 1954, 1974, 1988, 1998, and 2017, have been catastrophic, causing widespread damage, loss of life, and displacing millions of people. The Floods of 1954, 1955, 1974, 1983, 1987, 1988, 1998, 1999, 2003, 2004, 2007, 2014, 2016, 2017, 2019, 2020, 2021, 2022, and 2024 were highly destructive, with considerable casualties [14, 27, 28]. More significantly, floods engulfed 61%, 68%, 38% and 30% of the entire Bangladesh in 1988, 1998, 2004, and 2017, respectively, resulting in substantial loss equivalent to 4–10% (over two million metric tons) in rice production [29–31]. 57.01% households were exposed to floods with financial repercussions of US $0.85 billion from 2009 to 2014, while the 2017 flood impacted six million people, resulting in the deaths of 134 people [32]. The entire Rangpur division was inundated in the 2017 flood due to prolonged rainfall in the upstream of the Brahmaputra River, for which over 3000 people were left inaccessible in the char (sand bar) region of Gaibandha Sadar, Fulchhari, and Saghatta upazila, resulting in the inundation of 1,69,335 hectares of land and devastation of Aman rice of 5.5 lakh tons [33]. The estimated loss was about $ 140 million in Saghatta Upazila alone [34], and a lot of people were displaced, facing the shortage of food and pure drinking water, resulting in famine and waterborne disease [35]. Agricultural activity was severely disrupted, leading to economic destruction of the small-scale farmers [33, 36]. Poor communities face catastrophic consequences as floods in Gaibandha become a regular phenomenon [37].

Gaibandha, in northwest Bangladesh, is especially vulnerable to floods because of its close proximity to the Brahmaputra River and its low-lying topography. Gaibandha was severely submerged during the 1954, 1974, 1978, 1998, and 2019 floods [37]. For reducing the adverse socioeconomic damage of flood, a robust and comprehensive approach is indispensable, requiring proper planning, mitigation approach, forecasting, emergency preparedness, and implementation of preventive measures [38]. This research

gap emphasizes the necessity of a comprehensive flood susceptibility investigation study tailored to Gaibandha's unique hydrological and geographical conditions. By addressing this gap, this study aims to create a detailed flood susceptibility map for Gaibandha using three widely recognized MCDM techniques: Analytical Hierarchy Process (AHP), Analytic Network Process (ANP), and Fuzzy AHP (FAHP) [18, 39–45]. These techniques will allow for a thorough evaluation of various hydro-geomorphological factors, including elevation, slope, drainage density, rainfall, river proximity, and land use and land cover, in determining flood susceptibility in the district.

Decision-making complications have been widely resolved using weighted criteria based on expert judgement [15], denoted by AHP, ANP, and Fuzzy logic. Numerous studies are conducted on AHP, ANP, and FAHP globally for flood analysis, for instance, studies like [46–49] applied in critical sub-watershed mapping for flood assessment. According to research [49], ANP (0.79) outperformed AHP (0.77), while another one [46] demonstrates sensitivity analysis associated with flood susceptibility mapping validated by FEMA data, and a study [47] exhibits the inventory-based approach, where susceptible zone identification was emphasized. In Bangladesh, a few studies have applied, namely, flood hazard mapping of the Sangu River basin [49], flood susceptibility evaluation in the Brahmaputra River basin [50], flood effect on the livelihood in Gaibandha district [28], climate change impact in the Saghatta union of Gaibandha [51], flood mitigation and mapping approach in the Ghaghat River [52], flood susceptibility assessment in Gaibandha [34, 37] river erosion documentation based on AHP, damage and loss estimation due to floods using Synthetic Aperture Radar data [53], flood risk assessment based on MCDM [14], a flood study using geoinformatics application in Gaibandha, and a modeling approach in the Teesta river basin [54], GIS-based AHP for flood risk and susceptibility assessment, as demonstrated by [55] where rainfall (22%) and elevation (21%) emerge as the most dominant criteria, and lowest significance goes to TWI (1.5%), depicting a higher accuracy of 82.4% of AHP. According to the literature review, flood susceptibility mapping employing a combination of AHP, ANP, and FAHP was not performed in Gaibandha District before this study. Moreover, this region is understudied. Previous research in this region was only AHP-focused as a minor unit of comprehensive assessment. This study is the foremost application of AHP, ANP, and FAHP and compares them, establishing not only methodological but also spatial novelty. Because it is impossible to prevent flood risk in the downstream region, flood hazard assessment is essential to confirming strong and sustainable economic growth in this region [38, 56–58].

The main objective of this research is to create a reliable flood susceptibility map for Gaibandha and compare the effectiveness of the three MCDM approaches (AHP, ANP, and FAHP). To bridge all the research gaps, some holistic objectives have been designed, like (a) identifying strongly influencing hydro-geomorphological factors, (b) developing flood susceptibility maps for Gaibandha, (c) formulating and delineating flood-prone areas, and (d) comparing the level of susceptibility at the local or sub-district level through AHP, ANP, and FAHP approaches. By accomplishing this, the study presents a comprehensive and creative framework for assessing flood susceptibility, providing insightful information for emergency planning, policy-making, and flood prevention. The findings will help inform local and national strategies to reduce flood susceptibility, enhance preparedness, and promote resilience in flood-prone communities.

Additionally, the method used in the research may be applied to other areas with equivalent flood hazards, offering a globally applicable structure for disaster resilience and flood management associated with flood zoning, an updated early warning system, and adequate flood shelter.

2 Materials and methods

2.1 Study area

The tectonic-basinal extension of Gaibandha is between 25° 02′ N and 25° 39′ N latitude and 89° 11′ E and 89° 46′ E longitude on the foredeep of the Bengal basin, which is a floodplain tract associated with the Brahmaputra-Jamuna-Teesta braided river system [59, 60]. Gaibandha (Fig. 1) is one of the administrative units functioning as a district of the division of Rangpur of north-west Bangladesh, including 7 sub-districts, 82 unions, and 1250 villages. This district features a network of five rivers with a combined length of 107.71 km. This is encircled by the Teesta River; Chilmari upazila is situated in Kurigram district on the northern border, Piragachha of Rangpur district borders on the northeast, Mithapukur and Pirganj upazilas of the Rangpur region on the west, Ghoraghat upazila is part of Dinajpur district to the east, Kalai upazila of the Joypurhat district on the western side, and the southern portion of the Brahmaputra River flows side by side, with a total area of 2169.28 km^2 and a population of about 25,62,233 [61].

The climatic condition of Gaibandha district is characterized by moderate temperature, high humidity, and substantial rainfall [62]. This region experiences high temperatures of around ~33 °C during the summer season, extending from April to June; on the other hand, low temperatures of around ~26 °C during the winter season, extending from early November to late February, and experiences cold waves due to its proximity to the Himalayan hills [63]. The average annual temperature ranges from 11.2 to 32.3° C, and humidity levels are very high during the monsoon and sustain above normal conditions until termination of December [64]. Regarding the national average rainfall of ~2000 to 3000 mm, Gaibandha experiences a mean annual precipitation of about ~ 490 mm, which is more significant in the monsoon period from early June to the termination of September [65]. Varying slope from 0 to 47°, this fluvio-deltaic region has 3 types of

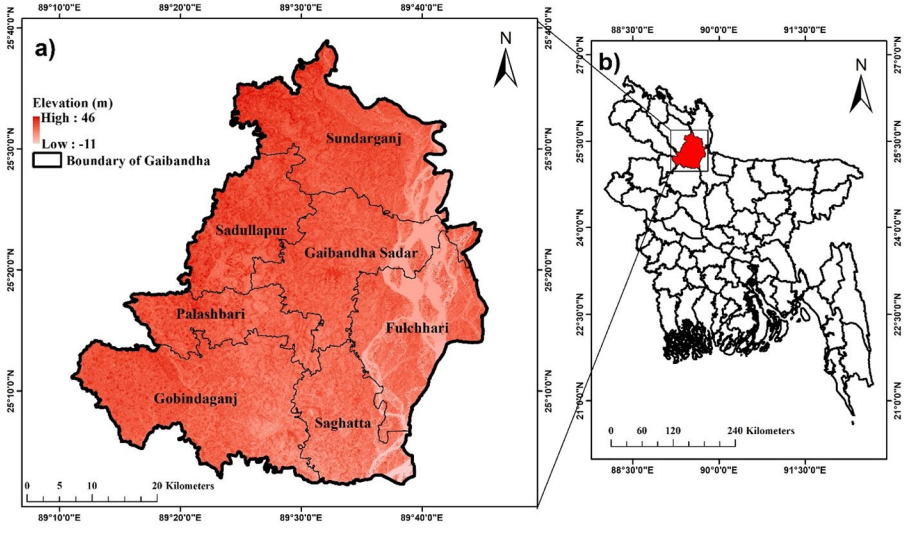


Fig. 1 Location map of the study area: **a** Gaibandha district; and **b** Gaibandha district in Bangladesh

soils (sandy, loamy, and clay), most households are made of mud, low-cost brick, bamboo, etc., making the socio-economic condition more vulnerable than the national average and susceptible to flooding. Water level of Brahmaputra touches its dangerous level between July-August, combined with the peak water level of Ganges between August-September, creating the most devastating flood situation and exposing 87% of households involved in primary economic activities as 85% out of 23,79,255 of total population are rural people, including agricultural assets like paddy, wheat, maize, mustard, watermelon, etc., which leads to huge economic loss every year [66, 67].

The topography, with varying slopes and soils ranging from sandy to clay, coupled with the region's low-lying nature, further exacerbates flood risks. Gaibandha's economy, largely based on agriculture, faces devastating losses every year due to flooding, particularly impacting rice production. Given the frequency and severity of floods, Gaibandha presents a critical case for studying flood susceptibility and improving flood management strategies, making it an ideal location for this research. This study involves the flood susceptibility assessment of the whole Gaibandha district, including 7 upazilas of the North-west region.

Figure 2 The schematic diagram outlines the complete methodological process that was followed in this study.

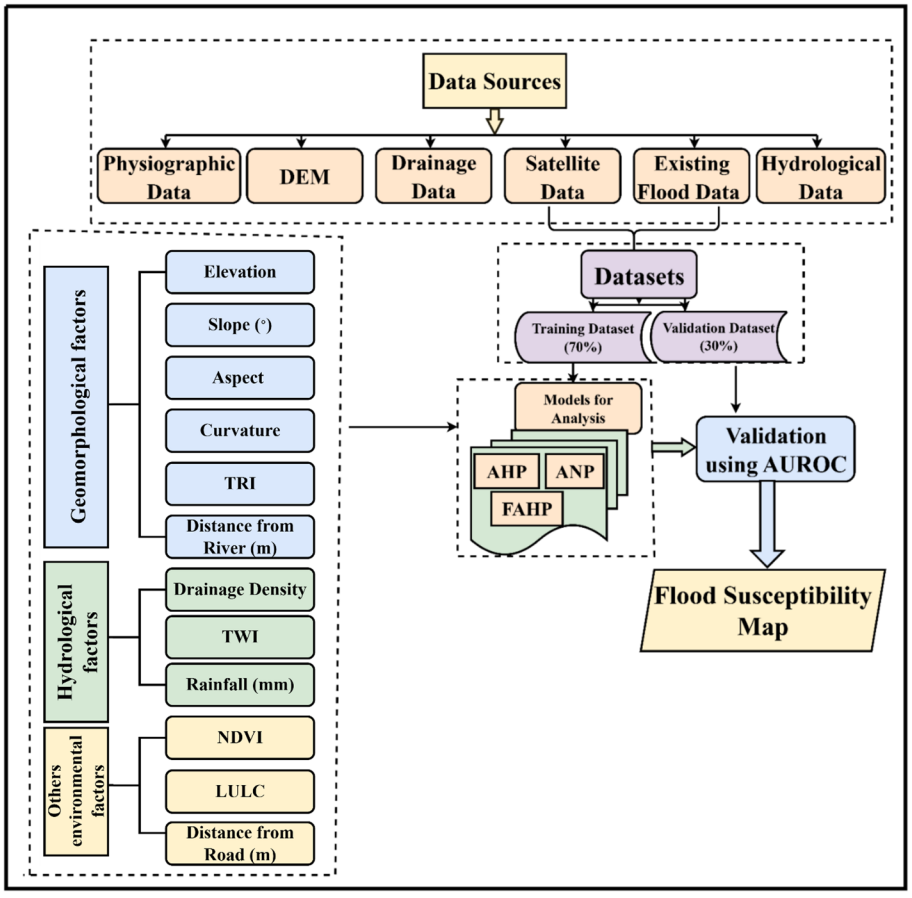


Fig. 2 depicts the methodological workflow followed in this study

2.2 Data collection

The most recent available and reliable data have been used, where topography is temporally stable, while other dynamic factors were exhibited employing updated data. The data used to conduct this study are displayed in Table 1.

2.3 Preparation of factor maps

The identification of the twelve flood-causing factors was shaped by a comprehensive review of related and applicable literature, the availability of data, and pronounced flood evaluation frameworks [46]. These factors span hydrological, geological, and other dimensions, which are cross-verified through expert judgements ensuring relevance to local geomorphological properties of Gaibandha District. The DEM was employed to formulate elevation, slope, aspect, curvature, TRI, and TWI, while a satellite image was prepared for LULC and NDVI. Other factors, like proximity to the river and road was generated from vector datasets. All the factor maps were produced using ArcGIS software 10.8 and collected data projected into WGS 1984 UTM Zone 45 N and resampled to a uniform resolution of 30 m.

2.3.1 Elevation

It is a physical component; the variations in elevation reflect the variation of vertical topographical appearance, therefore playing a significant part in creating large-scale floods [68]. In this study, the elevation raster map is divided into five sub-segments, indicating numerical ranking wherein the lowest elevation class denotes the highest susceptibility to flood risk and the highest elevation indicates low inundation (Fig. 3a).

2.3.2 Slope (degree)

It is a pivotal hydrologic factor. Steeper slopes accelerate overflow of surface runoff, increase the velocity of water, enhance soil erosion and vertical percolation rate, lessening the infiltration and interception rate over time, amplifying peak flood discharge rate. Consequently, gentle regions are more prone to experiencing massive flooding as

Table 1 Documentation of the secondary data sources used for a specific purpose

Data	Resolution	Data source	Utilization purpose	Year of data acquisition
Digital Elevation Model (DEM)	30 m	Shuttle Radar Topography Mission (SRTM)	Preparation of elevation, slope, aspect, curvature, TWI, TRI, and drainage density map	2020
Satellite image	30 m	Landsat 8 OLI/TIRS, USGS (Earth Explorer)	LULC classification and NDVI map preparation	2024
Rainfall data	0.5° (~ 50 km grid cell)	Climatic Research Unit, United Kingdom (https://crudata.uea.ac.uk/cru/data/hrg/)	Construction of an annual distribution map	2021–2024
Distance from the river	–	Humanitarian Data Exchange, HDX (https://data.humdata.org/dataset/bangladesh)	Stream network map preparation	2025
Distance from the road	–		Road network map preparation	2025
Sentinel-1 A Dual-Pol GRD High Resolution (VV & VH)	10 m	ASF Vertex (NASA)	Flood extent mapping and flood inundation analysis, preparation of inventory datasets	2024

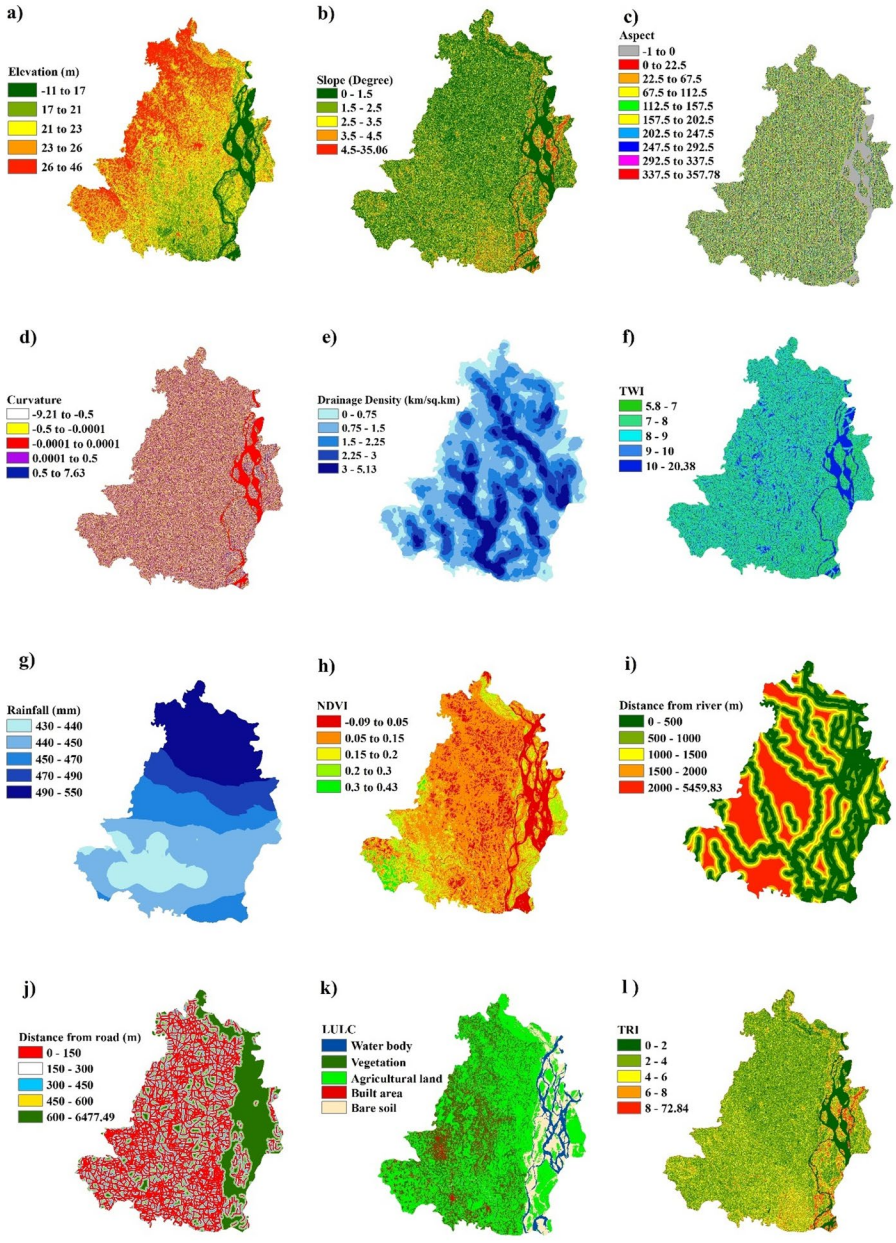


Fig. 3 Factors of flood susceptibility: **a** Elevation, **b** Slope, **c** Aspect, **d** Curvature, **e** Drainage density, **f** TWI, **g** Rainfall, **h** NDVI, **i** Distance from river, **j** Distance from road, **k** LULC, and **l** TRI

stagnant water accumulates and drains so slowly. The slope (degree) map (Fig. 3b) has been divided into five sub-groups.

2.3.3 Aspect

It usually relates to the horizontal direction that the mountain is facing [49]. This component affects hydrological parameters through evapotranspiration and the path of continental precipitation, determining the positioning of the slope, which is determined by the location of the ground surface [38]. The raster surface of aspect is categorized into ten classes (Fig. 3c).

2.3.4 Curvature

It is the single-direction rate of change in slope in a predetermined orientation [69]. Curvature explains sharpness as convex, concave, and flat, where concavity is positive, negative, and zero. Positive curvature accounts for increased runoff, while negative curvature depicts runoff retention [70]. Basin curvature of the Brahmaputra River along Gaibandha is concave up to flat, indicating a low gradient depositional floodplain. The curvature is subdivided into 5 categories (Fig. 3d).

2.3.5 Drainage density

Higher drainage density, influencing significant runoff and exceeding the channel capacity, contributes to severe and elevated flooding during prolonged rainfall [14, 38]. The drainage density map is divided into five categories using Eq. (1) (Fig. 3e).

$$D = \frac{L}{A} \tag{1}$$

where D is the Drainage density of the study area, L is the length of drainage channels (km), and A is the basin area (km^2) [14].

2.3.6 Topographic wetness index (TWI)

It is one of the significant factors of flood susceptibility [14]. This geographical measure of the water content of the watershed of a river basin can be presented mathematically in Eq. (2).

$$\mathrm{TWI} = \ln \frac{\alpha}{\tan \beta \ + C} \tag{2}$$

where α is the flow raster; β is the slope gradient in degree. TWI is plotted for the research area (Fig. 3f) and classified into five categories.

2.3.7 Rainfall

Intense and prolonged rainfall amplifies the likelihood and severity of river flow regimes and thereby intensifies the overflow tendency of the river when the river channel is unable to transmit surplus water [38]. Gaibandha experiences rainfall intensity ranging from 430 to 550 mm exceptionally elevated precipitation. Point data of rainfall raster map collected from "https://crudata.uea.ac.uk/cru/data/hrg/cru_cl_1.1/" and then classi fied into five categories (Fig. 3g).

2.3.8 Normalized difference vegetation index (NDVI)

Areas characterized by higher NDVI values represent dense vegetation, contributing to runoff control and a lower likelihood of flood susceptibility. An NDVI raster map was prepared through the analysis of multispectral imagery derived from the Landsat 8 satellite Operational Land Imager (OLI) sensor by using Eq. (3) (Fig. 3h). NDVI values range from − 1 to + 1 [71, 72].

$$NDVI = \frac{NIR - RED}{NIR + RED} \tag{3}$$

where *NIR* is the reflectance of near-infrared red band (band 5); *RED* is thereflectance of red band (band 4).

2.3.9 Distance from the river

The proximity to rivers is essential in terms of flood risks. The distance from the river raster map was divided into five categories (Fig. 3i). Therefore, the flood susceptibility possesses a significant influence on the minimal distance areas from the channel network. On the other hand, the likelihood of flood-susceptibility minimizes as the distance increases [49, 73, 74].

2.3.10 Distance from road

Proximity to road networks plays a critical hydrological factor influencing accessibility during flood hazard, determining susceptibility, emergency response, and recovery [75], thereby making evacuation routes and alternative access approaches, or government planning for exposed communities, more convenient [14]. Distance from the road is classified into five categories (Fig. 3j).

2.3.11 Land use and land cover (LULC)

Patterns of LULC strongly influence climatic hazards [76], including flood events; for example, built areas have less permeability and infiltration capacity, amplifying flood susceptibility [77]. Conversely, densely vegetated areas decrease flood frequency by de-amplifying runoff. Employing a satellite image (multispectral) derived from the Landsat 8 satellite's Operational Land Imager sensor, the LULC raster map was classified into five classes: waterbody, built-area, vegetation, agricultural land, and bare soil (Fig. 3k).

2.3.12 Topographic roughness index (TRI)

It describes the terrain's ability to reflect local elevation variability by quantifying the degree of surface roughness and topographic heterogeneity within a defined neighborhood [78]. High TRI, indicating greater surface irregularity, can impede surface runoff, thereby abating flood frequency. On the contrary, low TRI representing flat terrain leads to higher flood inundation. The TRI raster map is delineated into five sub-categories using Eq. (4) (Fig. 3l).

$$TRI = \frac{Area_v - Area_I}{Area_P - Area_I} \tag{4}$$

where $Area_v$ is the surface area, $Area_I$ is the flat area, $Area_P$ is the potential area.

2.4 Multicollinearity of the flood conditioning factors

The correlation between flood conditioning factors in a regression model can be described by multicollinearity, determining the interdependency, where a large relationship illustrates issues like model fitness and results interpretation [79]. Multicollinearity analysis of the factors is very important to ensure unbiasedness for determining the accuracy of the conditioning factors, eliminating the interdependency, and suggesting statistical significance [79, 80]. Factors with elevated multicollinearity must be overlooked [81, 82]. Variance inflation factors (VIF) and tolerance (TOL) are formulated using Eq. (6) and Eq. (5) for multicollinearity [79].

$$\mathrm{TOL} = 1 - R^2 \tag{5}$$

$$\mathrm{VIF} = \frac{1}{TOF} \tag{6}$$

where R^2 is the coefficient of determination of regression of the variable of interest onto other independent variables (flood conditioning factors).

VIF > 5 and TOL < 0.1 demonstrate disengagement of the multicollinearity problem [79].

Pearson's correlation coefficient evaluates the linear correlation between factors, where values greater than 0.5 indicates extraordinary level of collinearity employing Eq. (7) [82, 83].

$$r = \frac{\sum (X_i - \bar{X})\left((Y_i - \bar{Y}\right)}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}} \tag{7}$$

where r is the correlation coefficient, X_i and Y_i are the values for X and Y for the ith individual, $\bar{X}$ and $\bar{Y}$ are the means for X and Y.

The Pearson correlation and VIF values for the flood conditioning factors are shown in Figs. 4 and 5, respectively.

2.5 Analytical hierarchy process (AHP)

AHP, which is one of the widely applied forms of MCDM integrated with GIS (Geographic Information System) for assessing flood likelihood, was formulated by Thomas

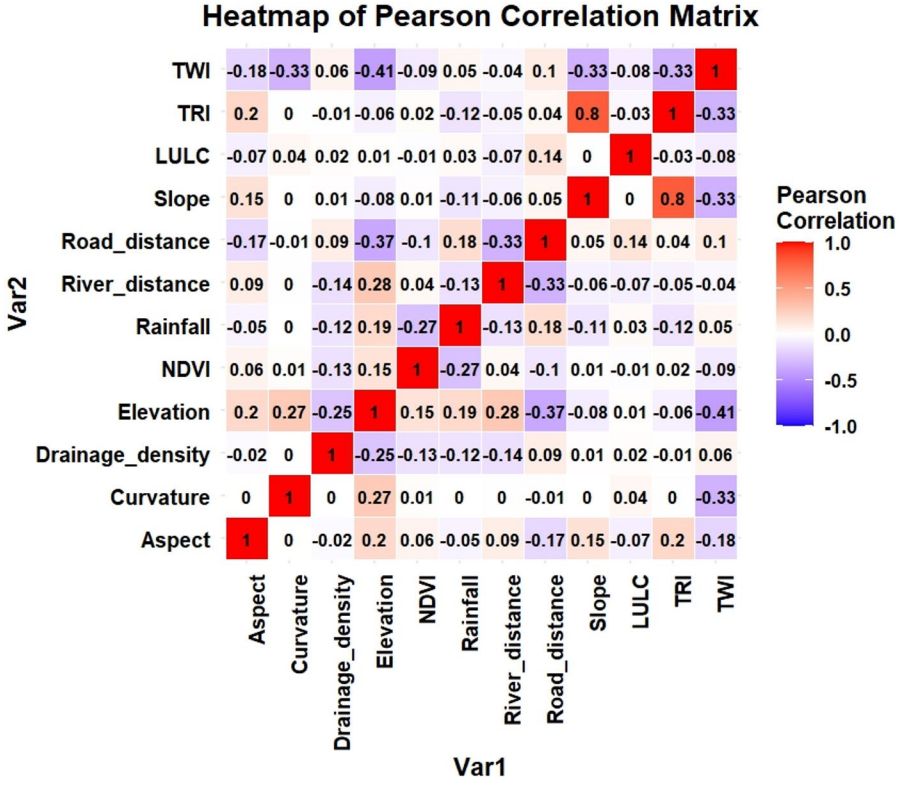


Fig. 4 Heat map of Pearson correlation coefficients between the flood conditioning factors

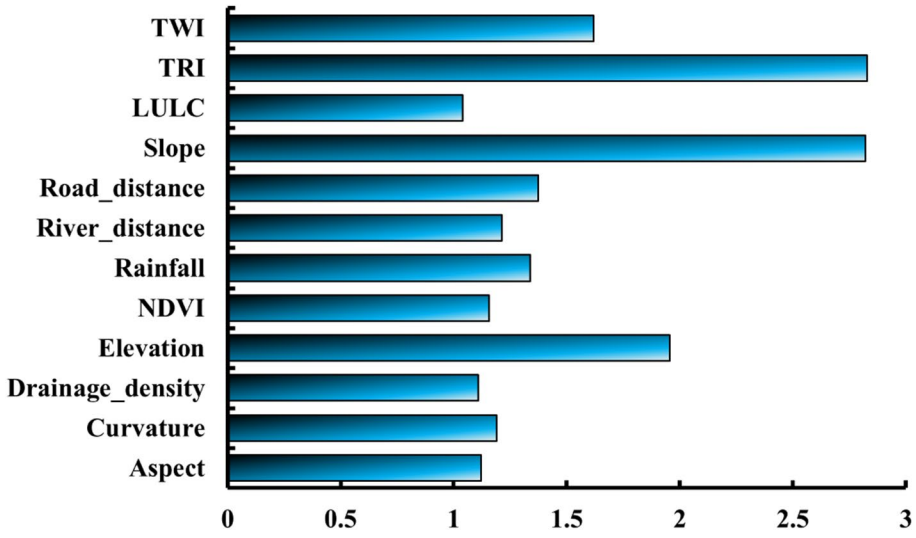


Fig. 5 Variance Inflation Factor (VIF) values of the flood conditioning factors

Table 2 Nine-point Saaty ranking scale used for AHP evaluation [87]

Significance	Significance	Explanation
1	Equally	Equal Influence of both factors
3	Moderate	Prefers one factor slightly over another
5	Strong	Prefers one factor strongly over another
7	Very strong	One parameter is strongly upheld
9	Extreme	One parameter is considered superior to another
2,4,6,8	Intermediate value	Boundaries are considered to be of critical importance

L.Saaty [68, 84, 85]. AHP also defines decision-making challenges through a systematic ranking of multiple criteria using a set of indicators [86]. Workflow of the AHP model-.

- Breaking down an unstructured problem into its prime components [38].
- Every factor and subgroup is weighted based on expert opinion using the Eigenvalue (Table 2) [87].
- The Consistency Ratio (CR) is formulated by Eq. (8).

$$\text{Consistency ratio (CR)} = \frac{\text{Consistency Index (CI)}}{\text{Random Index (RI)}} = \frac{\frac{\lambda_{\max}-\text{n}}{\text{n}-1}}{\text{RI}} \tag{8}$$

wherein λ is the maximum eigenvalue; n is the number of flood factors, and RI is the random consistency index.

The pairwise matrix should be readjusted when the CR value is above 0.1, mentioning the inconsistency [38] and perfect model consistency is achieved when its value equals zero [66]. In Table 3, RI values are listed for different n values.

Table 3 Random Index (RI) values corresponding to different matrix sizes or conditioning factors (n) [87]

n	1	2	3	4	5	6	7	8	9	10	11	12	13	...
RI	0.00	0.00	0.52	0.89	1.11	1.25	1.35	1.40	1.45	1.49	1.52	1.54	1.56	.

2.6 Analytic network process (ANP)

ANP, an advanced extension of the traditional AHP method, is used to overcome the inability [87, 88] developed by Saaty in 1996 [89]. Consequently, ANP allows both alternatives and criteria to be developed in relation to one another by facilitating comprehensive prioritization and it accounts for the interdependencies and feedback among variables exhibited in Fig. 6 [38]. The ANP process includes the following steps [19, 90]:

- Convert problems into a network structure [91] by conducting a pairwise matrix.
- Weighting data has to be tabulated based on expert opinion using the Geoman value.
- Local priority vector is computed using Eq. (9) [92]. Resulting weights formulate a Super matrix [93] using Eq. (10).

$$A = \lambda_{max} \tag{9}$$

where A is the pairwise comparison matrix; λ_{max} is the matrix's maximum value = $\sum_{j=1}^{n} \left[W_j \sum_{i=1}^{m} a_{ij} \right]$.

where a_{ij} is the value of factor; W_j is the weight

$$\mathrm{W} = \begin{pmatrix} {W_{i1}}^{(j1)} & {W_{i1}}^{(j2)} & \dots & {W_{i1}}^{(jn_j)} \\ {W_{i2}}^{(j1)} & {W_{i2}}^{(j2)} & \dots & {W_{i2}}^{(jn_j)} \\ \vdots & \vdots & \vdots & \vdots \\ {W_{in_i}}^{(j1)} & {W_{in_i}}^{(j2)} & \dots & {W_{in_i}}^{(jn_j)} \end{pmatrix}; \; W_{ij} = \text{component of super matrix} \tag{10}$$

- Identifying the final weight where CR $\leq$ 0.1 indicates Compatibility of the matrix estimated using Eq. (8)

2.7 Fuzzy analytical hierarchy process (FAHP)

FAHP is employed to estimate the weights of the criteria in the decision-making process, involving some steps in the application [95].

- A pairwise comparison matrix using Trapezoidal fuzzy scales (Table 4) represents fuzzy comparative judgements.
- CR and aggregation are performed (through the geometric mean) using Eqs. (8) and (11) by Buckley [96].

$$\mathrm{C} = \begin{pmatrix} C_{11} & C_{12} & \dots & C_{1m} \\ C_{21} & C_{22} & \dots & C_{2m} \\ \vdots & \vdots & \vdots & \vdots \\ C_{m1} & C_{m2} & \dots & C_{mm} \end{pmatrix} \tag{11}$$

$$Z_j = \left[\prod_{k=1}^{m} C_{jk} \right]^{1/m}; \quad \text{for j} = 1, 2, \dots, \text{m}; \quad \text{k} = 1, 2, \dots, \text{m}$$

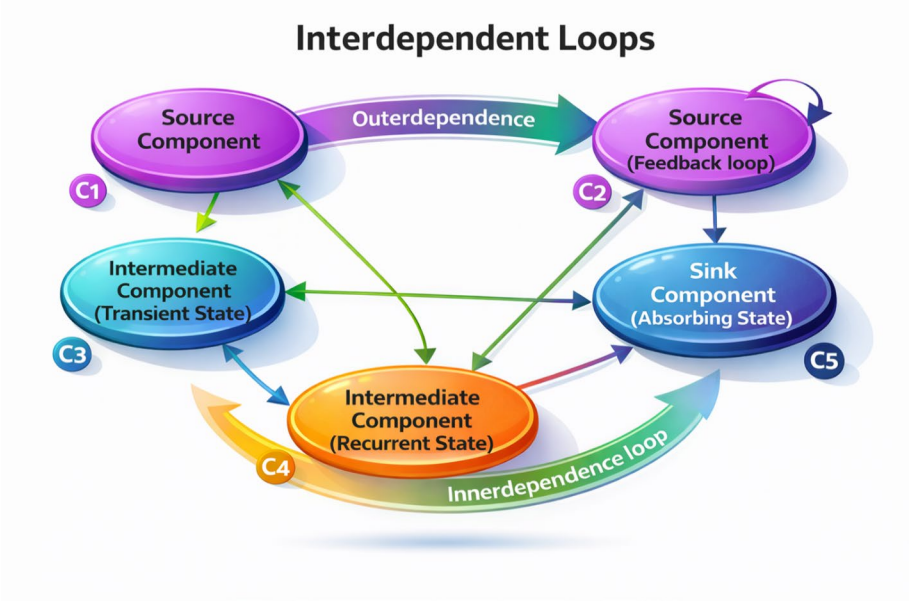


Fig. 6 Interdependent loops of the flood conditioning factors in the ANP model [94]

Table 4 Fuzzy evaluation scale [99]

Linguistic terms	Fuzzy score	Defuzzification value
Extremely strong	5/2, 3,7/2,4	3.250
Highly strong	2,5/2, 3,7/2	2.750
Moderately strong	3/2, 2,5/2, 3	2.250
Mildly strong	1, 3/2, 2,5/2	1.750
Balanced	1,1,1,1	1.000
Mildly weak	5/2,1/2,2/3,1	0.622
Moderately weak	1/3,2/5, 1/2,2/3	0.467
Highly weak	2/7, 1/3,2/5, 1/2	0.375
Extremely weak	1/4,2/7, 1/3,2/5	0.315

- The Extent Analysis method helps to address uncertainty [97, 98]. Sensitivity analysis tests the framework's stability [95].

The weights were diagnosed based on expert judgement, incorporating five specialists, where two geologists, two hydrologists, and one disaster management expert contributed with professional experience and knowledge of local topography and disaster dynamics. The weight assigning procedures for the susceptibility factors and their subclasses are shown in Tables 5 and 6.

2.8 Flood susceptibility mapping

Weights from AHP, ANP, and FAHP models were employed through a nine-point comparison scale, along with the percentage of factors' influence for flood susceptibility mapping that is performed using ArcGIS 10.8. The expression (Eq. (12)) was used to formulate the flood susceptibility index (*FSI*) map.

Table 5 A pairwise comparison matrix for the specified variables and respective weights for AHP, ANP, and FAHP models

Factors	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII	AHP weight (%)	ANP weight (%)	FAHP weight(%)	CR (%)
I	1	3	7	6	4	4	1/3	7	1	6	5	7	17.7	16.86	18.4	8.964
II	1/3	1	4	4	3	3	1/4	3	1/3	3	3	5	9.6	7.81	9.84	
III	1/7	1/4	1	1	1/4	1/3	1/6	1	1/6	1/2	1/2	2	2.4	2.69	2.72	
IV	1/6	1/4	1	1	1/4	1/4	1/6	1/2	1/5	1/2	1/3	1	2.2	2.81	2.27	
V	1/4	1/3	4	4	1	3	1/4	4	1/3	4	3	5	8.6	6.61	8.4	
VI	1/4	1/3	3	4	1/3	1	1/5	4	1/5	3	3	4	6.5	4.75	6.16	
VII	3	4	6	6	4	5	1	6	1/3	5	6	7	20.3	20.78	20.1	
VIII	1/7	1/3	1	2	1/4	1/4	1/6	1	1/6	2	2	2	3.5	3.10	3.3	
IX	1	3	6	5	3	5	3	6	1	6	5	6	20.3	25.56	20	
X	1/6	1/3	2	2	1/4	1/3	1/5	1/2	1/6	1	1/3	2	3.0	3.03	2.96	
XI	1/5	1/3	2	3	1/3	1/3	1/6	1/2	1/5	3	1	3	4.1	3.61	3.94	
XII	1/7	1/5	1/2	1	1/5	1/4	1/7	1/2	1/6	1/2	1/3	1	1.9	2.38	1.95	

I = Elevation, II = Slope, III = Aspect, IV = Curvature, V = Drainage density, VI = TWI, VII = Rainfall, VIII = NDVI, IX = Distance from river, X = Distance from road, XI = LULC, XII = TRI

$$FSI = (w_{Slope} \times Slope) + (w_{Elevation} \times Elevation) + (w_{Rainfall} \times Rainfall) + (w_{TRI} \times TRI) + (w_{NDVI} \times NDVI) + (w_{Distance\ from\ river} \times Distance\ from\ river) + (w_{Distance\ from\ road} \times Distance\ from\ road) + (w_{LULC} \times LULC) + (w_{TWI} \times TWI) + (w_{Curvature} \times Curvature) + + (w_{Aspect} \times Aspect) + (w_{Drainage\ density} \times Drainage\ density) \quad (12)$$

2.9 Validation of the flood susceptibility index map

Predictive performance assessment is a crucial approach for assessing the reliability of flood susceptibility results [100] employing the ROC (Receiver Operating Characteristic) curve [27, 101] and the AUC (Area under the Curve) value. The foremost aim of ROC curve validation is to delineate flood-exposed areas and examine non-flood-prone areas [102]. Associated angles with True Positive Rates (TPR) and False Positive Rates (FPR) are formulated through a total sample of 1509, signified as training data (70%) and testing data (30%). A higher angle, near 90°, demonstrates better results, indicating high TPR and low FPR using Eqs. (13) and (14) [103].

$$TPR = \frac{TP}{TP + FN} \quad (13)$$

$$FPR = \frac{FP}{FP + TN} \quad (14)$$

where *TP* is the True positives, *FN* is the False Negatives, *FP* is the False positives, *TN* is the True Negatives.

3 Results

3.1 Multicollinearity of the flood conditioning factors

The multicollinearity doesn't exist when Pearson's correlation matrix between flood conditioning factors is less than 0.6 [79]. TRI and slope had a positive correlation of 0.8, indicating the significance of multicollinearity. In contrast, a Pearson correlation value less than 0.6 illustrates the weak relationship between independent variables depicting using a Heat map of Pearson's correlation matrix in Fig. 4 [79]. Variance Inflation Factor (VIF) ranges from 1.04 to 2.83 (Fig. 5), indicating that the multicollinearity outputs are in accepted threshold of 5, demonstrating the nonexistence of multicollinearity problems. The low VIF values for aspect (1.12), curvature (1.19), drainage density (1.11), NDVI (1.15), LULC (1.04), Distance from river (1.2), and road (1.4) reveal that these conditioning factors result in predominantly independent relationships. The highest and lowest VIF were observed in TRI and LULC; hence, the model is free from multicollinearity concerns.

3.2 Flood susceptibility mapping

Final flood susceptibility maps were prepared by using weights derived from AHP, ANP, and FAHP-based factors associated with a weighted overlay analysis; therefore, the maps were categorized into five classes: very low, low, moderate, high, and very high (Fig. 7). In accordance with the AHP susceptibility map, very low susceptible area covers 140.522 km², but only 6.48% of the region is deemed to be at low susceptibility. As the susceptibility value increases to moderate, the affected area grows significantly, with 1,179.63 km² or 54.4% of the total area, classified under moderate susceptibility. While the very high susceptibility areas are critical, they cover only 24.1 km², which is

Table 6 Pairwise comparison matrix for the subclasses of each factor and determined weights for AHP, ANP, and FAHP models

	Subclasses	Sub-classes										CR (%)	AHP weight (%)	ANP weight (%)	FAHP weight (%)
		I	II	III	IV	V	VI	VII	VIII	IX	X				
Elevation (m)	**I**	**1**	3	5	7	8						6.71	50.4	55.17	50.38
	II	1/3	**1**	3	5	6							26.0	21.84	26.41
	III	1/5	1/3	**1**	3	4							13.2	11.05	12.89
	IV	1/7	1/5	1/3	**1**	2							6.2	6.60	6.11
	V	1/8	1/6	1/4	1/2	**1**							4.2	5.35	4.21
I = −11 to 17, II = 17 to 21, III = 21 to 23, IV = 23 to 26, V = 26 to 46															
Slope (Degree)	**I**	1	2	4	6	8						1.56	46.8	48.97	45.5
	II	1/2	1	2	4	6							26.8	25.52	27.5
	III	1/4	1/2	1	2	4							14.4	13.00	14.7
	IV	1/6	1/4	1/2	1	2							7.6	7.50	7.8
	V	1/8	1/6	1/4	1/2	1							4.4	5.02	4.5
I = 0 to 1.5, II = 1.5 to 2.5, III = 2.5 to 3.5, IV = 3.5 to 4.5, V = 4.5 to 35.063778															
Curvature	**I**	1	3	4	5	6						8.26	46.2	50.89	46.3
	II	1/3	1	3	4	5							26.1	21.90	26.5
	III	1/4	1/3	1	3	4							15.2	12.75	14.7
	IV	1/5	1/4	1/3	1	2							7.6	8.16	7.5
	V	1/6	1/5	1/4	1/2	1							5.0	6.29	5.0
I = −9.212611198 to −0.5, II = −0.5 to −0.0001, III = −0.0001 to 0.0001, IV = 0.0001 to 0.5, V = 0.5 to 7.628377															
Drainage density	**I**	1	3	4	5	6						8.26	46.2	50.90	46.3
	II	1/3	1	3	4	5							26.1	21.90	26.5
	III	1/4	1/3	1	3	4							15.2	12.75	14.7
	IV	1/5	1/4	1/3	1	2							7.6	8.16	7.5
	V	1/6	1/5	1/4	1/2	1							5.0	6.29	5.0
I = 0 to 0.75, II = 0.75 to 1.5, III = 1.5 to 2.25, IV = 2.25 to 3, V = 3 to 5.129318															

Table 6 (continued)

	Subclasses	Sub-classes										CR (%)	AHP weight (%)	ANP weight (%)	FAHP weight (%)
		I	II	III	IV	V	VI	VII	VIII	IX	X				
TWI	**I**	1	1/2	1/4	1/6	1/8						1.56	4.4	5.02	4.5
	II	2	1	1/2	1/4	1/6							7.6	7.50	7.8
	III	4	2	1	1/2	1/4							14.4	13.00	14.7
	IV	6	4	2	1	1/2							26.8	25.52	27.5
	V	8	6	4	2	1							46.8	48.97	45.5
I = 5.803437 to 7, II = 7 to 8, III = 8 to 9, IV = 9 to 10, V = 10 to 20.381151															
Rainfall (mm)	**I**	1	1/2	1/4	1/6	1/8						1.56	4.4	5.02	4.5
	II	2	1	1/2	1/4	1/6							7.6	7.50	7.8
	III	4	2	1	1/2	1/4							14.4	13.00	14.7
	IV	6	4	2	1	1/2							26.8	25.52	27.5
	V	8	6	4	2	1							46.8	48.97	45.5
I = 0.00043 to 0.00044, II = 0.00044 to 0.00045, III = 0.00045 to 0.00047, IV = 0.00047 to 0.00049, V = 0.00049 to 0.00055															
NDVI	**I**	1	2	4	6	8						1.56	46.8	48.97	45.5
	II	1/2	1	2	4	6							26.8	25.52	27.5
	III	1/4	1/2	1	2	4							14.4	13.00	14.7
	IV	1/6	1/4	1/2	1	2							7.6	7.50	7.8
	V	1/8	1/6	1/4	1/2	1							4.4	5.02	4.5
I = −0.093115 to 0.05, II = 0.05 to 0.15, III = 0.15 to 0.2, IV = 0.2 to 0.3, V = 0.3 to 0.434818															
Distance from River (m)	**I**	1	2	4	6	8						1.56	46.8	48.97	45.5
	II	1/2	1	2	4	6							26.8	25.52	27.5
	III	1/4	1/2	1	2	4							14.4	13.00	14.7
	IV	1/6	1/4	1/2	1	2							7.6	7.50	7.8
	V	1/8	1/6	1/4	1/2	1							4.4	5.02	4.5
I = 0 to 500, II = 500 to 1000, III = 1000 to 1500, IV = 1500 to 2000, V = 2000 to 5459.834961															

Table 6 (continued)

	Subclasses	Sub-classes										CR (%)	AHP weight (%)	ANP weight (%)	FAHP weight (%)
		I	II	III	IV	V	VI	VII	VIII	IX	X				
Distance from Road (m)	**I**	1	2	4	6	8						1.56	46.8	48.97	45.5
	II	1/2	1	2	4	6							26.8	25.52	27.5
	III	1/4	1/2	1	2	4							14.4	13.00	14.7
	IV	1/6	1/4	1/2	1	2							7.6	7.50	7.8
	V	1/8	1/6	1/4	1/2	1							4.4	5.02	4.5
I = < 4.5, II = 4.5–6, III = 6.01–7.5, IV = 7.501 – 10, V = > 10															
Aspect	**I**	1	1/2	1/3	1/4	1/3	2	3	4	5	5	9.26	10.9	9.23	10.0
	II	2	1	1/2	1/3	1/3	2	3	4	5	5		12.3	10.77	12.4
	III	3	2	1	1/2	1/2	1	2	3	5	5		13.1	12.45	14.5
	IV	4	3	2	1	1/2	1/3	2	3	5	5		15.9	15.61	15.3
	V	3	3	2	2	1	2	3	4	5	5		20.3	22.41	21.2
	VI	1/2	1/2	1	3	1/2	1	2	3	3	4		12.2	13.83	10.8
	VII	1/3	1/3	1/2	1/2	1/3	1/2	1	2	2	3		5.8	5.82	5.9
	VIII	1/4	1/4	1/3	1/3	1/4	1/3	1/2	1	2	3		4.2	4.10	4.4
	IX	1/5	1/5	1/5	1/5	1/5	1/3	1/2	1/2	1	2		2.9	3.05	3.1
	X	1/5	1/5	1/5	1/5	1/5	1/4	1/3	1/3	1/2	1		2.3	2.73	2.4
I = Flat (– 1), II = North (0–22.5), III = Northeast (22.5–67.5), IV = East (67.5–112.5), V = Southeast (112.5–157.5), VI = South (157.5–202.5), VII = Southwest (202.5–247.5), VIII = West (247.5–292.5), IX = Northwest (292.5–337.5), X = North (337.5–357.782196)															
LULC	**I**	1	2	4	4	4						9.05	40.9	44.02	40.8
	II	1/2	1	3	4	3							27.4	25.45	28.8
	III	1/4	1/3	1	3	1/3							10.7	10.15	9.8
	IV	1/4	1/4	1/3	1	1/2							6.6	8.09	6.7
	V	1/4	1/3	3	2	1							14.3	12.29	13.9
I = Water, II = Built area, III = Agricultural land, IV = Vegetation, V = Bare soil															

Table 6 (continued)

	Subclasses	Sub-classes										CR (%)	AHP weight (%)	ANP weight (%)	FAHP weight (%)
		I	II	III	IV	V	VI	VII	VIII	IX	X				
TRI	**I**	1	1/2	1/4	1/6	1/8						1.56	4.4	5.02	4.5
	II	2	1	1/2	1/4	1/6							7.6	7.50	7.8
	III	4	2	1	1/2	1/4							14.4	13.00	14.7
	IV	6	4	2	1	1/2							26.8	25.52	27.5
	V	8	6	4	2	1							46.8	48.97	45.5

I = 0 to 2, II = 2 to 4, III = 4 to 6, IV = 6 to 8, V = 8 to 72.839149

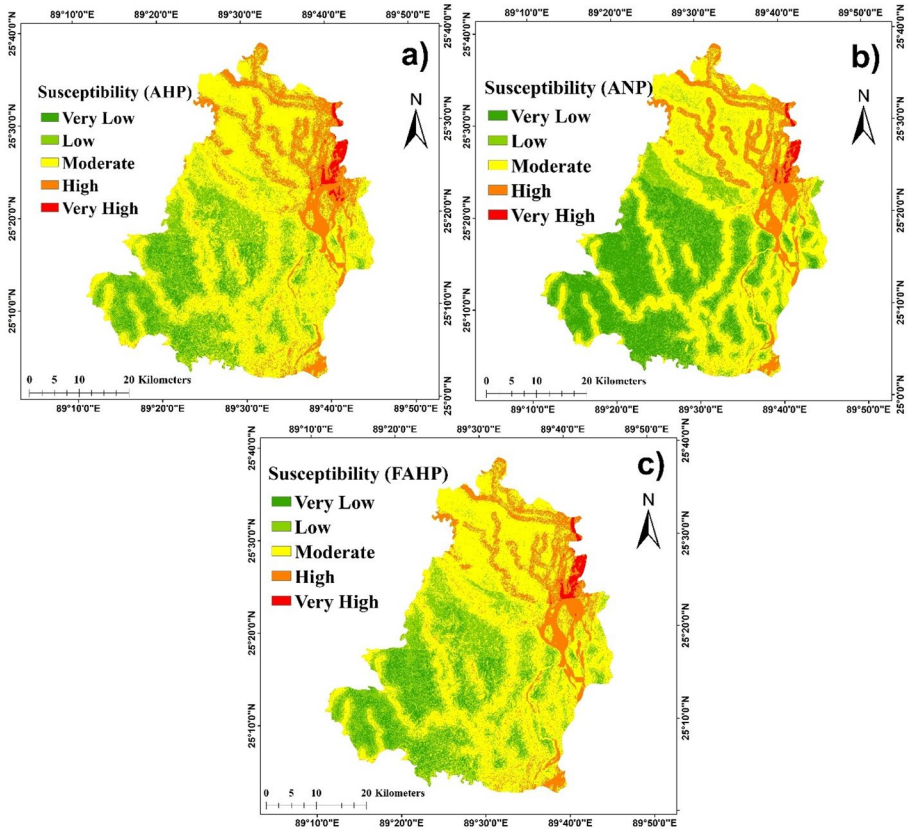


Fig. 7 Flood susceptibility maps of the study area based on **a** AHP, **b** ANP, and **c** FAHP models

just 1.11% of the area. Other key zones include the low-susceptible and high-susceptible areas, which make up 22% and 15.7%, respectively (Fig. 7a).

Shifting to ANP, a more detailed distribution of susceptibility can be observed. The very low susceptibility area now spans 431.714 km², and the affected area increases to 19.91%. This indicates a broader spread of susceptibility compared to AHP, as ANP considers the relationships between various factors. For moderate susceptibility, the affected area jumps to 43.70%, demonstrating that significant parts of the region are at moderate susceptibility. Even though the very high susceptibility zones are smaller, covering 14.8005 km², they still play a crucial role, with 0.68% of the area classified as extremely susceptible. Additionally, the low-susceptible and high-susceptible areas cover 23% and 12%, respectively (Fig. 7b).

In FAHP, we see a similar pattern, but with a more flexible approach due to the use of fuzzy logic. This allows FAHP to better account for uncertainty in the data, making it more adaptable. The area characterized as extremely low susceptibility is around 171.146 km² (8.15%), but FAHP places more emphasis on the higher susceptibility levels. The moderate susceptibility area is the largest, covering 1,146.42 km², or 52.87% of the region. The very high susceptibility areas, though small at 20.1753 km² (about 0.93%), are still important for targeting risk management efforts. Additionally, the low susceptibility and high susceptibility areas cover 25% and 13% of the region, respectively, showing a clear spread of varying risk levels across the area. (Fig. 7c).

The Southwest regions (Gabindaganj, Palashbari, some portion of Sadullapur, and Gaibandha Sadar) are low flood-prone areas, which is illustrated in Fig. 7. Contrarily, the

Eastern part (Fulchhari and some segments of Saghatta) and some portions of northern areas (a few fractions of Sadullapur, Gaibandha Sadar) are moderately flood susceptible, while the northern part (most subsets of Sundarganj and some portion of Fulchhari) exhibits high and very high flood susceptible conditions predominantly.

Ultimately, all three methods show that as susceptibility increases, the area affected by these zones also expands, especially in the moderate category. AHP offers a simpler and more static classification of susceptibility, whereas ANP accounts for interdependencies, resulting in a more balanced and evenly distributed risk map. FAHP, with its fuzzy logic, is the most adaptable, offering the most detailed view by capturing subtle variations in susceptibility. This makes FAHP especially useful in areas where the susceptibility assessment requires flexibility, uncertainty handling, and a more nuanced approach to understanding susceptibility.

The bar chart (Fig. 8) represents flood susceptibility levels across different upazilas using three methods: AHP, ANP, and FAHP. The upazilas displayed along the x-axis are Sundarganj, Sadullapur, Gaibandha Sadar, Fulchhari, Palashbari, Gabindaganj, and Saghatta. In AHP, Sundarganj and Sadullapur have large, very low susceptibility (green) areas, with some moderate susceptibility (yellow) zones. Gaibandha Sadar and Fulchhari show more moderate susceptibility (yellow), while Palashbari and Gabindaganj are mostly very low (green) and low (light green). ANP shows a more even spread of susceptibility. Fulchhari and Gabindaganj have larger moderate (yellow) and high susceptibility (orange) zones, with more moderate susceptibility in Gaibandha Sadar. Sundarganj also has moderate (yellow) areas, but high susceptibility (orange) zones are more widespread. FAHP, using fuzzy logic, highlights very high susceptibility (red) areas, especially in Sundarganj and Sadullapur. Gaibandha Sadar and Fulchhari show more moderate (yellow) and high susceptibility (orange), while Palashbari and Gabindaganj remain very low (green) to low (light green). Very high susceptibility (red) areas are minimal.

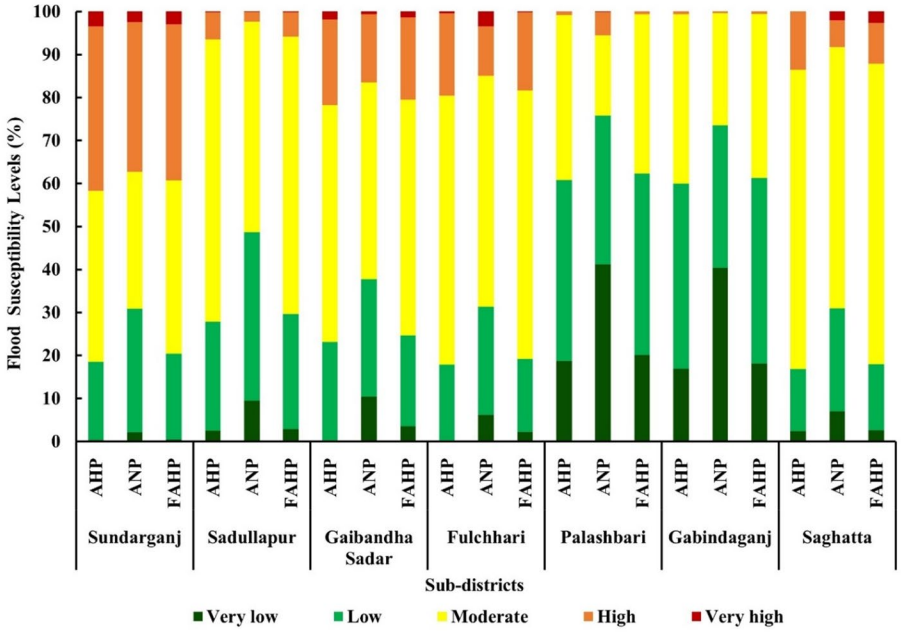


Fig. 8 Flood susceptibility levels (%) based on susceptible area of AHP, ANP, and FAHP models for all the upazilas (sub-districts) of the study area

Overall, FAHP provides the most detailed and accurate assessment of flood susceptibility, focusing on very high susceptibility areas, while AHP gives a simpler, more generalized susceptibility map. ANP offers a balanced approach, accounting for relationships between different factors, leading to a more evenly distributed susceptibility across the upazilas.

3.3 Validation of flood hazard map using the ROC curve

The models' performance was validated by the AUROC. The flood susceptibility maps of AHP, ANP, and FAHP have a higher AUC of 0.881, 0.885, and 0.883, respectively, as shown in Fig. 9, generated by the traditional approach, indicating that ANP slightly outperformed AHP and FAHP, with minimal difference. Therefore, the ANP model displayed an excellent performance in this study, having a higher AUC value, indicating comparative robustness in decision-making dynamics, while the AHP model exhibited the lowest AUC value, indicating comparative lower strength. Therefore, AHP demonstrates the least reliable model, while ANP appears as the best approach in the context of Gaibandha District based on the used datasets. On the other hand, the AUROC value decreased according to DeLong's test and bootstrapping, demonstrating an AUC value of 0.727, 0.739, 0.727 for AHP, FAHP, and ANP, respectively exhibiting in Fig. 10. This reduction is predicted as DeLong and bootstrap is responsible for sampling variability. In spite of numerical differences, all the model depicts good and acceptable performance, where FAHP was marginally superior and identical accuracy is evaluated for AHP and ANP. The AUC value of the model curve depicts greater precision of the analysis.

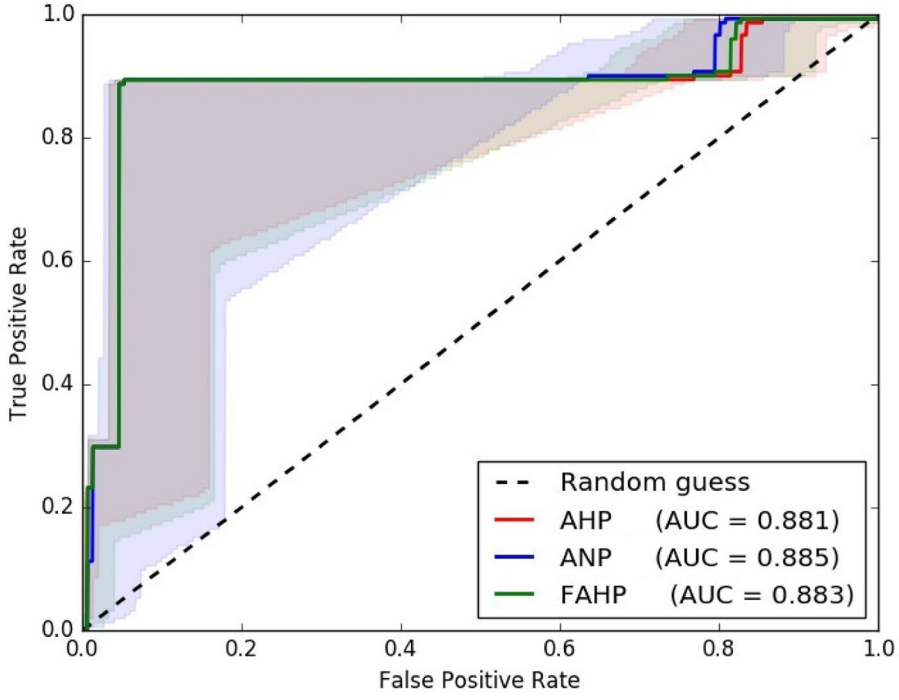


Fig. 9 AUC values and ROC curves for AHP, ANP, and FAHP models derived from ArcGIS 10.8

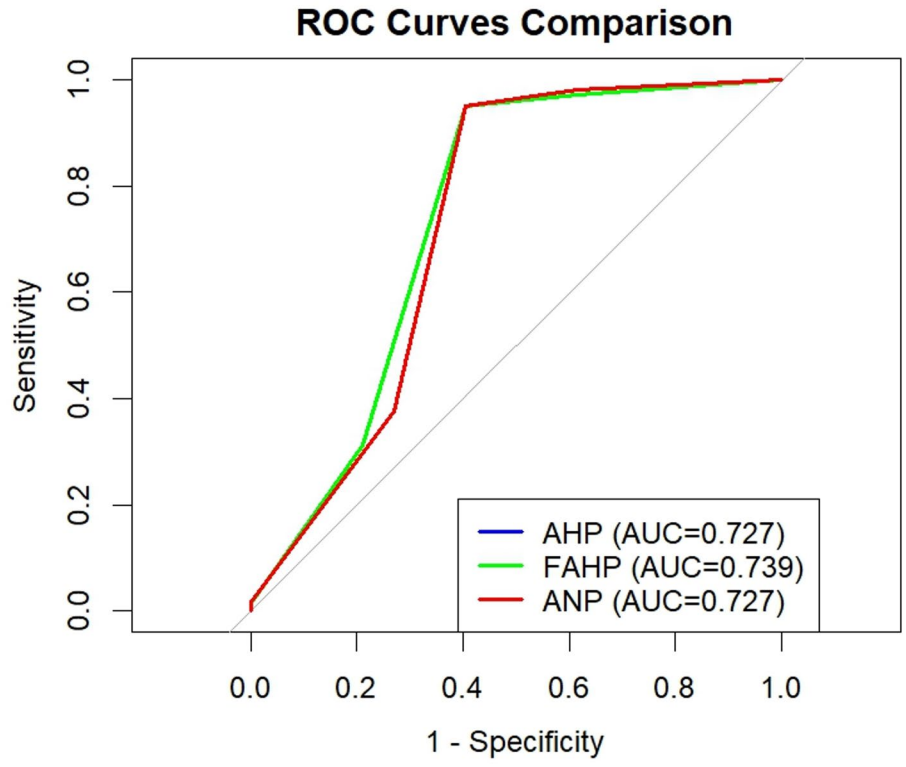


Fig. 10 AUC values and ROC curves for AHP, ANP, and FAHP models derived from DeLong's and bootstrapping test

4 Discussion

Gaibandha is one of the most flood-prone regions in Bangladesh, although it remains under-researched, largely due to a lack of comprehensive, high-resolution flood data. This data scarcity presents a significant challenge when trying to apply more traditional, data-intensive flood risk models. The small number of experts involved in this study may not fully capture the complexity of the flood susceptibility assessment for Gaibandha. A larger panel of experts with expertise in hydrology, geology, and disaster management would provide a more comprehensive perspective, especially for assigning weights to the flood-conditioning factors [104, 105]. As a result, we turned to expert-based MCDM methods (AHP, ANP, and FAHP) to assess flood susceptibility in this region based on the available expert opinions. These methods are particularly suitable for areas where reliable data is limited, and their flexibility allows for incorporating expert judgment into the modelling process. However, future studies should adopt a more systematic, data-driven approach when more reliable and comprehensive data, along with expert opinions, become available. Our findings align with previous studies that consistently identify rainfall and proximity to rivers as critical factors influencing flood susceptibility. For instance [55], highlighted the significance of rainfall and proximity to rivers in the Brahmaputra basin, while [106] observed similar findings in their study of the Teesta River basin. These consistent results underscore the fundamental importance of hydrological factors in flood susceptibility assessments across Bangladesh [26].

However, what sets our study apart is the use of AHP, FAHP, and ANP models, where ANP outperformed AHP and FAHP by considering the interdependencies between different susceptibility factors. While all three methods demonstrated identical accuracy,

the differences between them were marginal, with ANP achieving the highest accuracy of 0.885 (traditional ROC-AUC analysis), slightly outperforming both AHP (0.881) and FAHP (0.883). This result supports findings from [92, 107], who similarly found that ANP provided more accurate flood risk predictions compared to AHP and FAHP. On the other hand, FAHP slightly outperformed AHP and ANP, demonstrating the accuracy of 0.739, 0.727, and 0.727, respectively, in DeLong's and bootstrapping tests [4, 15, 86, 108].

Despite these promising results, several challenges and limitations emerged. A key limitation lies in the subjectivity of expert judgment during the weight assignment process, which can lead to inconsistencies across models. This subjectivity is well-documented in existing literature [38], emphasizing the need for a more data-driven, objective approach when sufficient data becomes available. Additionally, the reliance on medium-resolution satellite imagery, coupled with the lack of detailed flood history in Gaibandha, meant that some aspects of flood susceptibility mapping lacked the temporal precision necessary for more robust decision-making. This challenge has been addressed in previous research, where higher spatial resolution data led to improved susceptibility mapping accuracy [109].

Nevertheless, the flood susceptibility maps generated using AHP, ANP, and FAHP methods align with prior studies in Gaibandha, which have identified the northern and riverine areas as highly vulnerable to flooding [14, 53]. These regions, especially those near the Brahmaputra River, are frequently subjected to severe floods, particularly throughout the monsoon season. The flood susceptibility maps produced in this study help confirm the susceptibility of these regions and offer new insights into areas with varying degrees of flood risk, as noted in several studies [37, 54]. However, there is a need to further examine the magnitude and spatial patterns of susceptibility across Gaibandha to fully understand the impacts of its unique geomorphological and hydrological conditions. Looking ahead, as more hydrological data becomes available, such as high-resolution DEMs, rainfall data, and flood-event records, ANP and other models can be enhanced by machine learning or physically-based hydrological methods. These approaches could provide more reliable and accurate flood risk predictions, as demonstrated in several studies [86, 110, 111].

Overall, while the ANP and FAHP models performed the best in terms of accuracy, the choice of model should depend on the specific requirements of the flood susceptibility assessment. The differences between the models were marginal, and all may be acceptable tools for susceptibility mapping in Gaibandha District, particularly in situations where more sophisticated models are not feasible [112]. This study expands on the rapidly increasing body of information about the assessment of flood risks in Bangladesh and establishes a strong framework for future research. However, the data limitations emphasize the need for improved local data collection. As more comprehensive datasets become available, integrating machine learning and hybrid models will be key to improving the precision and reliability of flood risk assessments, helping policymakers and local authorities develop more effective flood management strategies [9, 113]. These advanced methods would offer more reliable and accurate flood risk predictions, addressing some of the limitations faced by current models. Additionally, the integration of local community knowledge and feedback could significantly improve the relevance and accuracy of flood susceptibility assessments, ensuring that the models better reflect local conditions

and needs. Future research should also focus on improving satellite imagery resolution, conducting sensitivity analysis of the MCDM weights, and addressing subjectivity in weight assignment to further improve flood susceptibility predictions and provide more effective and robust flood management strategies.

5 Conclusion

This study provides a comprehensive flood susceptibility analysis for Gaibandha District, Bangladesh, utilizing three widely recognized MCDM methods: AHP, ANP, and FAHP. The analysis revealed key factors influencing flood-prone areas, with rainfall and proximity to rivers consistently identified as the primary risk factors, aligning with previous studies in Bangladesh. Notably, the use of AHP, ANP, and FAHP models provided valuable insights into the strengths and weaknesses of each approach. Among the three methods, the ANP model demonstrated the highest accuracy, with an AUC value of 0.885, surpassing both AHP (0.881) and FAHP (0.883). The differences between the models were marginal, but ANP's ability to consider interdependencies between different flood-conditioning factors contributed to its superior performance. FAHP showed a slight edge over AHP and ANP in DeLong's and bootstrapping tests, with AUC values of 0.739, 0.727, and 0.727, respectively, indicating its higher accuracy in handling uncertainty and variability.

The findings suggest that flood-prone areas in Gaibandha are predominantly influenced by hydrological factors, including rainfall and proximity to rivers, with significant implications for flood susceptibility management. These models provide useful tools for policymakers and local authorities, guiding decisions on flood preparedness, infrastructure development, and resource allocation. However, challenges such as the subjectivity of weight assignment, and the limited resolution of satellite imagery remain. These limitations highlight the need for more robust, data-driven approaches and the integration of higher-resolution data as it becomes available. The study makes several important recommendations for hydrologists and future research in flood susceptibility mapping:

- Expand the expert panel to include hydrology, geology, and disaster management specialists to enhance the reliability and accuracy of weight assignments.
- Integrate machine learning models and hybrid approaches to better handle large, complex datasets and improve prediction accuracy.
- Increase satellite imagery resolution and flood event data to enhance spatial and temporal precision in flood susceptibility maps.
- Incorporate local knowledge and community-based approaches to ensure models reflect local conditions and enhance relevance.

Despite the challenges and limitations, this research has contributed to enhancing our understanding of flood susceptibility mapping in Gaibandha, providing a comprehensive framework for future studies. The combination of MCDM methods, GIS, and remote sensing technologies presents a reliable approach to flood susceptibility analysis, with future improvements focused on integrating more precise data and advanced models. These advances are essential for building resilience against floods in Bangladesh and other similar regions, where effective flood risk management strategies are critical for protecting lives, livelihoods, and infrastructure.

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Author contributions
M.M.R. conceptualized the study and supervised the overall research. The methodology was designed by M.M.R., F.A.P., and S.J.E. Software development and data processing were carried out by M.M.R., F.A.P., and S.J.E. Validation of the result was performed by M.M.R. and A.T. Formal analysis was conducted by M.M.R., F.A.P., and A.T. Data curation was handled by M.M.R., A.T., F.A.P., and S.J.E. The original draft was prepared by F.A.P. and S.J.E., while review and editing were performed by M.M.R. and A.T. Visualization was prepared by M.M.R. and A.T. All authors have gone through and approved the finalized version of the article.

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The supporting data of this investigation are accessible upon request from the corresponding author.

Declarations

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